

APPENDIX C

a trading game



C.1 INTRODUCTION

A lot of people reading this book will never have traded options or even stocks. In this appendix I describe a very simple trading game so that a group of people can try out their skill without losing their shirts. The game is based on that by one of my ex-students David Epstein.

C.2 AIMS

The aims of this game are to familiarize students with the basic market-traded derivative contracts and to promote an understanding of the concepts involved in trading, such as bid, offer, arbitrage and liquidity.

C.3 OBJECT OF THE GAME

To make more money than your opponents. After the final round of trading, each player sums up their profits and losses. The player who has made the most profit is the winner.

C.4 RULES OF THE GAME

1. One person (possibly a lecturer) is the game organizer and in charge of choosing the types of contracts available for trading, the number and length of the trading rounds, judging any disputes and jollying the game along during slack periods.
2. The trading game takes place over a number of rounds. At the end of each round, a six-sided die is thrown. After the last round, the 'share price' is deemed to be the sum of all of the die rolls.
3. Traded contracts may include some or all of forwards, calls and puts at the discretion of the organizer. The organizer must also decide what exercise prices are available for call or put options.

4. All contracts expire at the end of the final round. The settlement value of each traded contract can then be determined by substituting the share price into the appropriate formula. A player's profits and losses on each trade can then be calculated and the resultant profit/loss is their final score.
5. During a round, players can offer to buy or sell any of the traded contracts. If another player chooses to take them up on their offer, then the deal is agreed and both parties must record the transaction on their trading sheet.
6. A deal on a contract must include the following information:
 - Forward: forward price and quantity
 - Call or put: type of option (call or put), exercise price, cost and quantity

The organizer chooses the types of contracts available and the strike prices.

The forward price or option cost and the quantity in a deal are chosen by the players.

For beginners, play three games in succession, with the following structures:

1. Play with just the forward contract.
2. Play with the forward contract and the call option with exercise price 15.
3. Play with the forward and the call and put options with exercise price 15.

All three games take place over five rounds, each five minutes in length.

C.5 **NOTES**

1. Depending on the level of prior knowledge of the players, the organizer may need to explain the characteristics of the various traded contracts. It will be instructive to emphasize that the forward contract has no cost initially.
2. There will probably be times when the organizer has to act as a 'market-maker' and promote trading, for instance, asking the group if anyone wants to buy shares or at what price someone is willing to do so.
3. For more advanced students, consider introducing some of the following ideas:
 - Increase the number of rounds
 - Decrease the length of each round
 - Include extra calls and puts with different exercise prices or which either come into existence or expire at different times. You must fix the details of these extra contracts in advance of the game
 - Include other contracts, e.g. Asian options or barriers
 - Include a second die for a second underlying share price

4. Including futures with 'daily' marking to market can be tried, but slows down the game. Nevertheless, it does illustrate the importance of margin, especially if the students have a limit on how much 'in debt' they are allowed to become.

C.6 HOW TO FILL IN YOUR TRADING SHEET

C.6.1 During a trading round

In the 'Contract' column, fill in the specifications of the instrument that you have bought/sold. Specify the forward price or exercise price if applicable (e.g. if there is more than one contract of this type in the game).

In the 'Buy/Sell' column, fill in whether you have bought or sold the contract and the quantity.

In the 'Cost per contract' column, fill in the cost of a single contract.

C.6.2 At the end of the game

In the 'Settlement value' column, fill in the value of a single contract with the final share price.

In the 'Profit/loss per contract' column, fill in the profit/loss for a single contract.

In the 'Total profit/loss' fill in the total profit/loss for the trade (= profit/loss $\times$ quantity).

Example During a round, your transactions are:

Buy 10 call options, with exercise price 20, at a cost of 2 each. Sell 1 put option, with exercise price 15, at a cost of \$1. Buy 5 forwards, with forward price 19.

Your trading sheet should be filled in as below.

Contract	Buy/sell	Cost per contract	Total cost	Settlement value	Profit/loss
Call 20	Buy 10	2			
Put 15	Sell 1	1			
Forward 19	Buy 5	–			

At the end of the game, the final share price is 21. Consequently, the trading sheet is completed as follows.

Contract	Buy/sell	Cost per contract	Total cost	Settlement value	Profit/loss
Call 20	Buy 10	2	$21 - 20 = 1$	$1 - 2 = -1$	$-1 \times 10 = -10$
Put 15	Sell 1	1	0	$1 - 0 = +1$	$+1 \times 1 = +1$
Forward 19	Buy 5	–	$21 - 19 = 2$	+2	$+2 \times 5 = +10$

The total profit and loss for the trader is therefore $-10 + 1 + 10 = +1$.

Remember that:

If you buy a contract, your profit/loss = settlement value – cost per contract

If you sell a contract, your profit/loss = cost per contract – settlement value

Trading sheet

[illegible]

The Trading Game—designed by David Epstein, 1999.

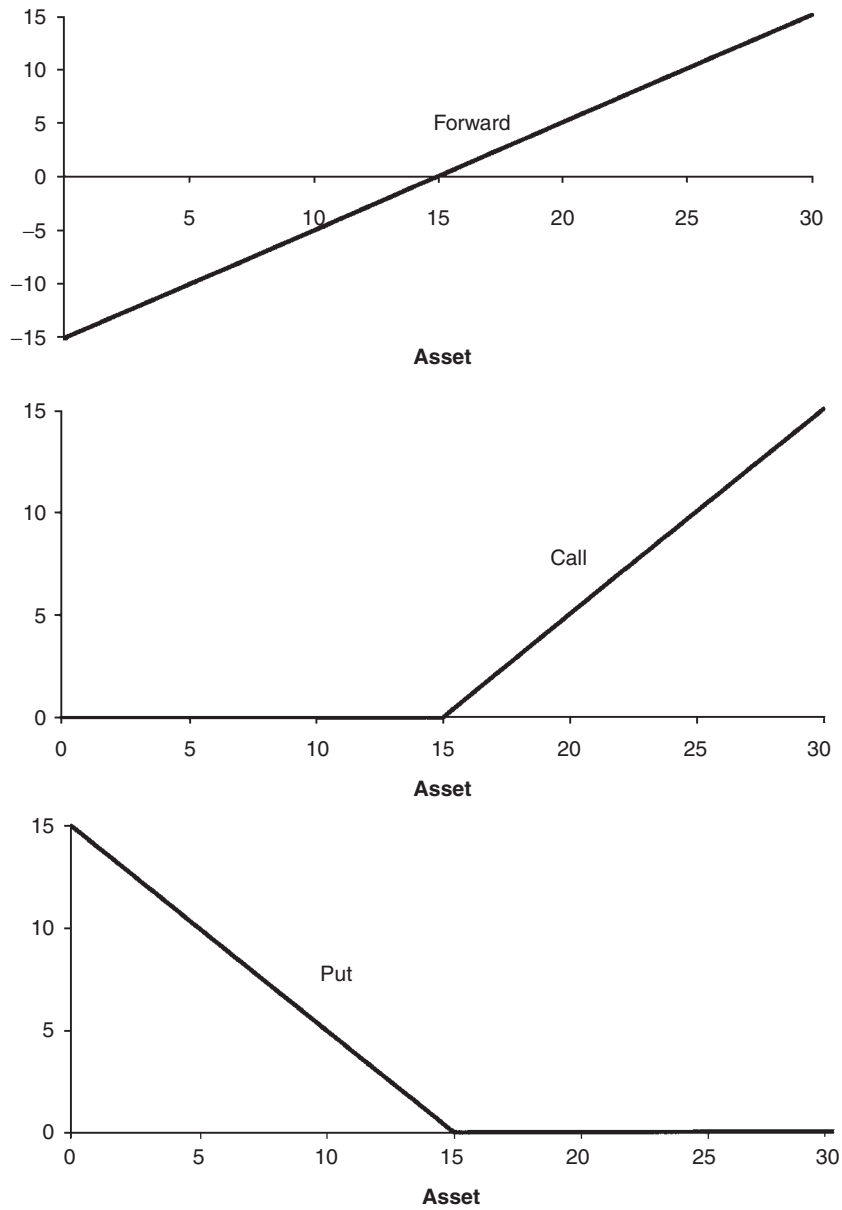


Figure C.1 Available contracts.

